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Teo Metrica Pseudohiperbolica Desigualdad Triangular Generalizada

Teorema 1 (Desigualdad triangular generalizada).

$$\forall z_1, z_2, z_3 \in \mathbb{D} : \varrho(z_1, z_2) \leqslant \frac{\varrho(z_1, z_3) + \varrho(z_2, z_3)}{1 + \varrho(z_1, z_3)\, \varrho(z_2, z_3)}.$$

***Demostración*:** Basta demostrar la desigualdad para $z_1' = \tau_{z_3}(z_1)$, $z_2' = \tau_{z_3}(z_2)$ y $z_3' = \tau_{z_3}(z_3) = 0$, ya que ϱ es invariante por τ_{z_3}. Como se tiene que

$$\varrho(z_1, z_2) = \varrho(z_1', z_2'), \quad \varrho(z_1, z_3) = \varrho(z_1', 0) = |z_1'|, \quad \varrho(z_2, z_3) = \varrho(0, z_2') = |z_2'|,$$

basta probar que $\varrho(z_1', z_2') \leqslant \frac{|z_1'|+|z_2'|}{1+|z_1'||z_2'|}$, que es consecuencia del `cor-borde-bola-pseudohiperbolica-des`

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Referenciado en

- Prop metrica poincare
- Teo Completitud Metrica Pseudohiperbolica

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